

Data Analytics: Visualisation and Programming - BAA1026

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Power BI Report

AgriFoods Global Analysis

Group 1

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Table of Contents

Company & Data Insight	3
Global Sales Overview	3
Profitability Analysis	4
Operational Efficiency	5
Sales Representative Insights	6
Business Strategies and Conclusion	7
Appendix.....	9

Company & Data Insight

AgriFoods Global is an international company operating across diverse markets. The company's data provides insights into sales performance, customer demographics, product categories, and regional operations. For this report, the dataset "powerbi_data_group1_AgriFoods_Global.xlsx" served as the foundation for developing an analytical dashboard in Power BI that highlight sales trends and operational efficiency across countries, products, and representatives.

Before importing the data into Power BI, the raw dataset was cleaned and prepared in Microsoft Excel. A unique Sales ID was created to ensure the data met at least First Normal Form (1NF). The Country and Category columns were deleted due to redundancy and overlap with other fields. Minor inconsistencies such as duplicate records, date formatting issues, and missing values were corrected to improve data quality. The cleaned and structured dataset, saved as "Sales.xlsx," was then used in Power BI to enable accurate visualisations and generate reliable insights into AgriFoods Global's sales performance. (See Appendix 1.1)

It is important to acknowledge certain data limitations. The analysis is based on historical records from 2015 to 2024. Additionally, the dataset lacks external economic or seasonal variables that could influence demand or cost trends, limiting conclusions to the internal business data available.

The analysis was developed through four report pages within the dashboard: Global Sales Overview, Profitability Analysis, Operational Efficiency and Sales Representative Insights. Each report page focuses on a distinct aspect of business performance, providing management with a clear, data-driven understanding of AgriFoods Global's operations and strategic outcomes.

Global Sales Overview

This section of the report will analyse how sales figures are broken down across the regions in which AgriFoods Global operates. When we look at revenue by region, we see that South America is the leader, closely followed by Africa and North America, then Asia, and finally Europe, which takes a significantly lower share of the profits. (See Appendix 2.1)

One of the main insights we can gather from this data is the poor performance in Africa when it comes to turning revenue into profits. Despite having the second-highest revenue regionally, it has generated the second-lowest profit, €4.28 million from a total revenue of €28.98 million

(see Appendix 2.2). This inefficiency is evident when we compare with the other regions. Despite generating less revenue, both North America and Asia have higher profits, with both regions hitting at least €6 million in profit. Europe also displays far greater efficiency in turning revenue into profit; with the lowest revenue of €21.97 million, it still managed to turn a profit of €3.36 million.

Sales channels are broken into online and offline. Online Sales generates slightly more revenue and profit, with €15.85 million in profit compared to €14.79 million. If we analyse this by region, we see that most follow a roughly even split of sales by channel, except in North America. Here, customers have shown a preference for online purchases (see Appendix 2.3), with about 28,000 units sold online compared to 20,000 offline. This accounts for about 62% of revenues, indicating a strong online presence and robust digital infrastructure in this region.

We can see AgriFoods' global reach and strong regional performance, especially in South America, North America, and Asia. However, our analysis indicates that adjustments are required in Europe and Africa to bring them up to the standards demonstrated in the other regions. Our global sales analysis highlights problems with Europe's revenue and Africa's profitability.

Profitability Analysis

AgriFoods Global generated €136.10M in revenue and €30.64M in profit from 251,000 units sold between Q1 2015 and Q3 2024 (see Appendix 3.1). Despite solid revenue, the company's Gross Profit Margin (23%) and Net Margin (6.12%) reveal a significant gap. This could indicate high operating costs and margin leakage. The Data consistently shows Tablets and Insurance Plans with negative margins. Laptops and Monitors dominate in profitability charts.

Gross Margin reflects product-level profitability before expenses, which is strong across most regions and products in AgriFoods Global. Net Margin accounts for operational costs, logistics and overhead, which is much lower and more volatile in this case. The 17-point gap between gross and net margin suggests inefficiencies in cost control, especially in regions like Africa, and products like Tablets and Printers.

South America emerged as the most profitable region, contributing €11.93 million in profit with an average net margin of 18.38% (Appendix 3.2). This success was driven by high-margin products, including Monitors (38.19%), Laptops (42.64%), and Smartphones (19.78%). In contrast, Africa, despite being the second-highest revenue generator, delivered only €4.28 million

in profit, with a net margin of just 2.51%. This inefficiency stems from losses in key products like Tablets (-23.48%), Printers (-25.99%), and Solar Panels (-14.44%).

Asia and North America demonstrated balanced profitability, each contributing over €6 million in profit. Asia's strength lay in Monitors (26.32%) and Solar Panels (21.47%), while North America benefited from strong online sales and high-margin products like Solar Panels (38.35%) and Tablets (23.49%). Europe, although generating the lowest revenue, demonstrated operational efficiency, converting €21.97 million in revenue into €3.36 million in profit, with standout margins in Flight Tickets (31.84%) and Building Materials (28.53%).

Time-series analysis across the report page highlights volatility in quarterly profits, with peaks in 2020 Q1 (€3M) and 2021 Q1 (€2.51M), which demonstrates periods of strong margin conversion. Profit troughs in 2016 Q1 (-€282K) and 2018 Q3 (-€514K) highlight periods where rising costs or inefficiencies eroded gross margin. The report page shows that net margin fluctuates more sharply than gross margin, underscoring the impact of operational factors over time. These fluctuations suggest external market factors and internal operational shifts that warrant further investigation. Channel analysis reveals a near-even split between online and offline sales, with online channels slightly outperforming in profit (€15.85M vs €14.79M). North America leads in digital adoption, with 62% of sales online, indicating a robust e-commerce infrastructure.

Based on this analysis, we can see that AgriFoods Global's profitability is driven by strategic product performance and regional efficiency.

Operational Efficiency

AgriFoods Global's operational efficiency varies significantly across products and is highlighted by the following metrics: Profit per Unit, Shipping Cost per Unit, Net Margin %, and Weighted Discount % (Appendix 4.1). These indicators reveal how effectively each product converts sales into profit while managing operational costs.

Laptop and Monitor emerge as top performers. Laptop (Appendix 4.2) shows a Profit per Unit of €130.92, a Net Margin % of 25.46%, a Weighted Discount % of 16.36%, and a moderate Shipping Cost of 0.06 per unit. Monitor (Appendix 4.3) also performs well, with a Profit per Unit of €38.19, Net Margin % of 7.37%, and Weighted Discount % of 16.43%, supported by

strong unit sales of 35,673. These products demonstrate consistent profitability and operational stability.

Solar Panel and Smartphone also show solid performance. Solar Panel has a Profit per Unit of €41, a Net Margin % of 7.61%, and a Weighted Discount % of 18.03%, while Smartphone delivers a Profit per Unit of €37.16, a Net Margin % of 6.48%, and a Weighted Discount % of 17.6%.

In contrast, Tablet, Flight Ticket, and Car underperform. Tablet (Appendix 4.4) shows a negative Profit per Unit of -€43.02, a Net Margin % of -8.09%, and a Weighted Discount % of 16.39%, with unit sales of 25,137. Flight Ticket and Car both report Net Margin % below 0.4%, with low profit per unit, €1.85 and €2.13 respectively, despite similar discount levels and shipping costs.

Building Material presents a strong case, with a Profit per Unit of €107.34, Net Margin % of 19.54%, and a relatively low Weighted Discount % of 13.56%, indicating efficient pricing and cost control.

This analysis highlights the wide variation in operational efficiency across AgriFoods Global's product portfolio. Products such as Laptop, Monitor, Solar Panel and Smartphone demonstrate strong profitability and margin resilience, supported by balanced discounting and manageable shipping costs. In contrast, underperforming items like Tablet, Flight Ticket and Car reveal structural inefficiencies, with low or negative margins and minimal profit per unit. Building Material stands out as high value, with efficient cost control and pricing strategy. Overall, the data highlights how different products contribute to the overall operational performance, with clear insight to where value is created and lost.

Sales Representative Insights

When examining total revenue by sales representative, Bob emerges as the top performer, closely followed by Charlie and Eve, then Diana, and finally Alice. This section interprets the Power BI Sales Representative report page, highlighting regional performance trends to inform strategic decision-making. This report page is illustrated in Appendix 5.1.

Sales performance across the five representatives reveals variation in both revenue generation and profitability. Bob stands out as the strongest performer, achieving one of the highest total

revenues (€28.24M) and contributing significantly to overall profit growth, as shown in Appendix 5.2. His results indicate strong pricing discipline and cost control, positioning him as a key driver of organisational profitability. Alice also performed well, generating a total revenue of €24.61M, reflecting consistent execution and stable performance across markets.

In comparison, Charlie and Diana achieved similarly high revenue levels, ranging from €27M to €28M, suggesting balanced contributions to total sales and net margins of 5.98% and 8.66%, respectively. However, their profitability outcomes appear less efficient than Bob's.

Eve recorded a negative net profit of –€130.24K despite achieving a total profit of €4.51M and total revenue of €27.74M, resulting in a net margin of –0.47% (see Appendix 5.3).

Overall, the Power BI Sales Representative report page highlights clear variations in sales and profitability performance across the team. Bob consistently outperforms other representatives, demonstrating strong revenue generation and efficient cost control, making him a key contributor to overall profitability. Alice also delivers stable and consistent results, reinforcing her role as a reliable performer. Charlie and Diana achieve competitive revenue levels but exhibit lower margins, indicating potential inefficiencies in their sales or pricing approaches. In contrast, Eve's negative net margin, despite solid revenue figures, suggests underlying profitability challenges. Collectively, these findings emphasise the importance of balancing revenue growth with profit efficiency to sustain strong organisational performance.

Business Strategies and Conclusion

The Power BI analysis of AgriFoods Global's performance identifies four core strategic areas essential to improving profitability, operational consistency, and long-term competitiveness. Each strategy directly addresses specific performance gaps revealed through the dashboard.

The *Digital Expansion Strategy* focuses on strengthening Europe's online presence. Europe demonstrates low total revenue despite effective cost control and high net margins, primarily due to limited online reach and weak digital infrastructure. By implementing North America's successful online sales framework, expanding e-commerce platforms. Through improving digital marketing and integrating regional customer data for targeted promotions, AgriFoods can

enhance online visibility and customer engagement. This approach would allow Europe to increase sales volume and revenue without significant additional fixed costs, narrowing the regional performance gap.

The *Cost Optimisation Strategy* aims to improve Africa's profitability. Although Africa generates the second-highest revenue, it suffers from one of the lowest net margins (2.51%), caused by high shipping costs and inefficient operations. A comprehensive cost review should be conducted, with a focus on logistics and supplier management. Introducing regional warehousing and diversifying suppliers can reduce shipping costs, while implementing cost-tracking KPIs will ensure sustained improvement. These measures would strengthen Africa's cost structure, increase its net margins, and turn the region into a more reliable profit contributor.

The *Product Portfolio Strategy* addresses unbalanced profitability across product categories. Several offerings, such as Tablets, Printers, and Insurance Plans, deliver persistent negative margins, weakening the company's overall profitability. AgriFoods should reallocate investments toward high-performing products such as Laptops, Monitors, Solar Panels, and Smartphones, while considering phasing out or redesigning low-margin items. Marketing should align with consumer channel behaviour, promoting offline-strong categories such as Building Materials separately from digital-first products like Tablets. This refined focus will improve overall margins, optimise capital allocation, and reduce exposure to underperforming segments.

Finally, the *Performance Development Strategy* targets disparities in individual sales performance. Bob consistently outperforms peers through strong pricing discipline and cost control, while others, such as Eve and Diana, underperform. Establishing a structured mentorship program based on Bob's practices, standardising discount policies, and linking sales KPIs to profitability will improve team consistency and efficiency.

In conclusion, these four integrated strategies (digital expansion, cost optimisation, product focus, and performance development) form a cohesive framework to make AgriFoods Global more data-driven, efficient, and profitable. Implementing them will enhance operational balance, strengthen margins, and position the company for sustainable global growth.

Appendix

Appendix 1.1

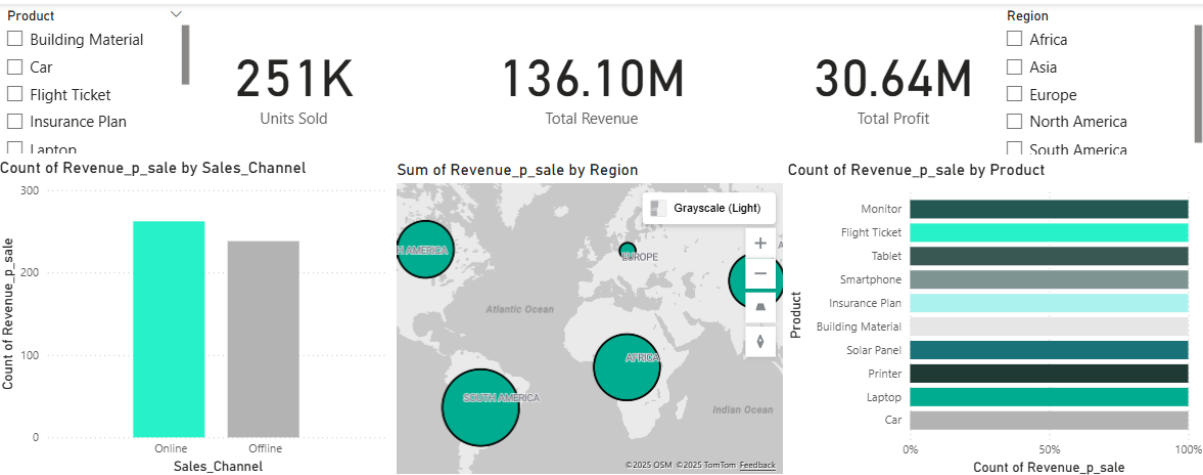
Organization	Date	Region	Country	Product	Category	Sales_Channel	Order_Priority	Units_Sold	Unit_Price	Unit_Cost	Customer_ID	Sales_Rep	Discount	Shipping_Cost	Revenue_p_sale	Total_Cost	Total_Profit	Net_Profit
AgriFoods C	#####	Asia	India	Solar Panel	Agriculture	Online	Medium	574	691,51	191,58	86406	Charlie	0,26	28,89	396926,7	109966,9	286959,8	183730
AgriFoods C	#####	Africa	Brazil	Car	Healthcare	Offline	Low	254	746,91	360,73	20675	Bob	0,29	19,35	189715,1	91625,42	98089,72	43052,98
AgriFoods C	#####	North Ame	USA	Solar Panel	Agriculture	Offline	Low	541	179,55	436,69	44639	Bob	0,02	7,53	97136,55	236249,3	-139113	-141063
AgriFoods C	#####	Africa	India	Tablet	Education	Online	Medium	334	303,15	621,99	64510	Bob	0,09	26,66	101252,1	207744,7	-106493	-115632
AgriFoods C	#####	Europe	South Afric	Insurance P	Healthcare	Online	High	976	387,19	779,01	52026	Alice	0,1	21,69	377897,4	760313,8	-382416	-420228
AgriFoods C	#####	North Ame	India	Tablet	Education	Online	Low	721	289,13	141,42	11936	Bob	0,25	24,09	208462,7	101963,8	106498,9	54359,14
AgriFoods C	#####	Asia	Germany	Tablet	Constructi	Offline	High	96	315,79	407,35	20529	Diana	0,08	44,34	30315,84	39105,6	-8789,76	-11259,4
AgriFoods C	#####	Europe	India	Monitor	Finance	Online	Medium	362	594,74	722,55	69940	Bob	0,16	44,55	215295,9	261563,1	-46267,2	-80759,1
AgriFoods C	#####	Europe	India	Solar Panel	Electronics	Offline	Medium	233	333,9	566,65	23027	Diana	0,18	11,68	77798,7	132029,5	-54230,8	-68246,2
AgriFoods C	#####	Europe	USA	Monitor	Finance	Offline	High	120	818,15	455,18	78203	Eve	0,27	8,79	98178	54621,6	43556,4	17039,55
AgriFoods C	#####	South Ame	Brazil	Solar Panel	Travel	Online	High	995	444,17	392,52	32691	Bob	0,29	8,88	441949,2	390557,4	51391,75	-76782,4
AgriFoods C	#####	South Ame	Brazil	Building M	Constructi	Offline	Medium	898	747,51	77,56	51199	Bob	0,09	16,04	671264	69648,88	601615,1	541185,3
AgriFoods C	#####	Africa	South Afric	Laptop	Constructi	Offline	High	26	990,33	317,84	62797	Charlie	0,15	33,86	25748,58	8263,84	17484,74	13588,59
AgriFoods C	#####	Asia	USA	Flight Ticke	Constructi	Offline	Low	114	295,21	324,73	59212	Charlie	0,17	28,45	33653,94	37019,22	-3655,28	-9114,9
AgriFoods C	#####	Asia	USA	Smartphon	Constructi	Offline	Low	599	899,97	298,6	80295	Charlie	0,21	38,67	539082	178861,4	360220,6	246974,7
AgriFoods C	#####	Asia	India	Building M	Travel	Offline	Low	219	460,99	133,74	95395	Alice	0,21	43,74	100956,8	29289,06	71667,75	50423,08
AgriFoods C	#####	Europe	India	Monitor	Automotiv	Offline	High	757	80,83	586,06	80978	Bob	0,06	13,34	61188,31	443647,4	-382459	-386144
AgriFoods C	#####	Asia	Germany	Building M	Travel	Offline	Low	471	532,51	141,36	88350	Eve	0,3	32,1	250812,2	66580,56	184231,7	108955,9
AgriFoods C	#####	Europe	Brazil	Flight Ticke	Energy	Online	Medium	113	258,2	188,51	19360	Diana	0,21	12,08	29176,6	21301,63	7874,97	1735,8
AgriFoods C	#####	South Ame	South Afric	Laptop	Energy	Online	Low	795	975,91	151,32	18523	Bob	0,17	29	775848,5	120299,4	655549,1	523625,8

Sales_Id	Date	Sales_Channel	Customer_ID	Region	Product	Order_Priority	Units_Sold	Unit_Price	Unit_Cost	Shipping_Cost	Revenue_p_sale	Cost_p_sale	Profit_p_sale	Net_Profit_p_sale	Sales_Rep	Discount
1	01.01.15	Online	86406	Asia	Solar Panel	Medium	574	691,51	191,58	28,89	396926,7	109966,9	286959,8	183730	Charlie	0,26
2	08.01.15	Offline	20675	Africa	Car	Low	254	746,91	360,73	19,35	189715,1	91625,42	98089,72	43052,98	Bob	0,29
3	15.01.15	Offline	44639	North America	Solar Panel	Low	541	179,55	436,69	7,53	97136,55	236249,3	-139112,74	-141063	Bob	0,02
4	22.01.15	Online	64510	Africa	Tablet	Medium	334	303,15	621,99	26,66	101252,1	207744,66	-106492,56	-115631,91	Bob	0,09
5	29.01.15	Online	52026	Europe	Insurance Plan	High	976	387,19	779,01	21,69	377897,44	760313,76	-382416,32	-420227,75	Alice	0,1
6	05.02.15	Online	11936	North America	Tablet	Low	721	289,13	141,42	24,09	208462,73	101963,82	106498,91	54359,14	Bob	0,25
7	12.02.15	Offline	20529	Asia	Tablet	High	96	315,79	407,35	44,34	30315,84	39105,6	-8789,76	-11259,37	Diana	0,08
8	19.02.15	Online	69940	Europe	Monitor	Medium	362	594,74	722,55	44,55	215295,88	261563,1	-46267,22	-80759,11	Bob	0,16
9	26.02.15	Offline	78203	Europe	Solar Panel	Medium	233	333,9	566,65	11,68	77798,7	132029,45	-54230,75	-68246,2	Diana	0,18
10	05.03.15	Online	78203	Europe	Monitor	High	120	818,15	455,18	8,79	98178	54621,6	43556,4	17039,55	Eve	0,27
11	12.03.15	Online	32691	South America	Solar Panel	High	995	444,17	392,52	8,88	441949,15	390557,4	51391,75	-76782,38	Bob	0,29
12	19.03.15	Online	51199	South America	Building Material	Medium	898	747,51	77,56	16,04	671263,98	69648,88	601615,1	541185,3	Bob	0,09
13	26.03.15	Offline	62797	Africa	Laptop	High	26	990,33	317,84	33,86	25748,58	8263,84	17484,74	13588,59	Charlie	0,15
14	02.04.15	Offline	59212	Asia	Flight Ticket	Low	114	295,21	324,73	28,45	33653,94	37019,22	-3655,28	-9114,9	Charlie	0,17
15	09.04.15	Offline	80295	Asia	Smartphone	Low	599	899,97	298,6	38,67	539082,03	178861,4	360220,63	246974,73	Charlie	0,21
16	16.04.15	Offline	95395	Asia	Building Material	Low	219	460,99	133,74	43,74	100956,81	29289,06	71667,75	50423,08	Alice	0,21
17	23.04.15	Offline	80978	Europe	Monitor	High	757	80,83	586,06	13,34	61188,31	443647,42	-382459,11	-386143,75	Bob	0,06
18	30.04.15	Offline	88350	Asia	Building Material	Low	471	532,51	141,36	32,1	250812,21	66580,56	184231,65	108955,9	Eve	0,3
19	07.05.15	Online	19360	Europe	Flight Ticket	Medium	113	258,2	188,51	12,08	29176,6	21301,63	7874,97	1735,8	Diana	0,21
20	14.05.15	Online	18523	South America	Laptop	Low	795	975,91	151,32	29	775848,45	120299,4	655549,05	523625,81	Bob	0,17

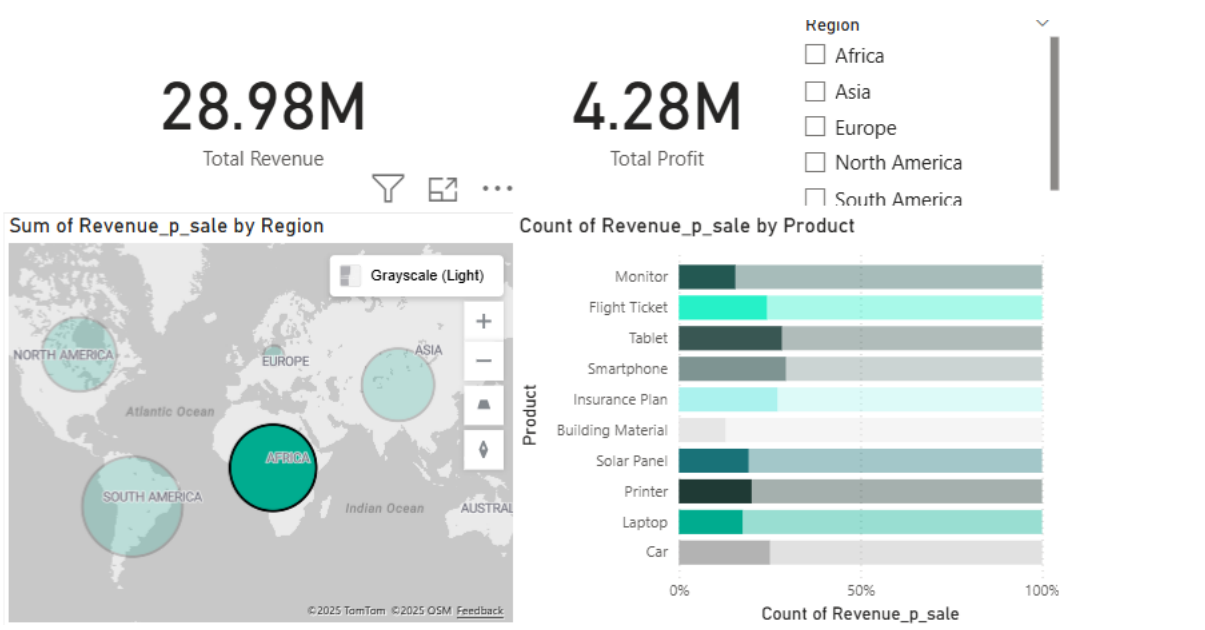
Appendix 2.1

The Dashboard can be accessed through the link for a more detailed overview.

Sales Dashboard 10.11.2025.pbix



Appendix 2.2



Sum of Revenue_p_sale by Region

Count of Revenue_p_sale by Product

Grayscale (Light)

NORTH AMERICA

SOUTH AMERICA

EUROPE

AFRICA

ASIA

AUSTRALIA

Atlantic Ocean

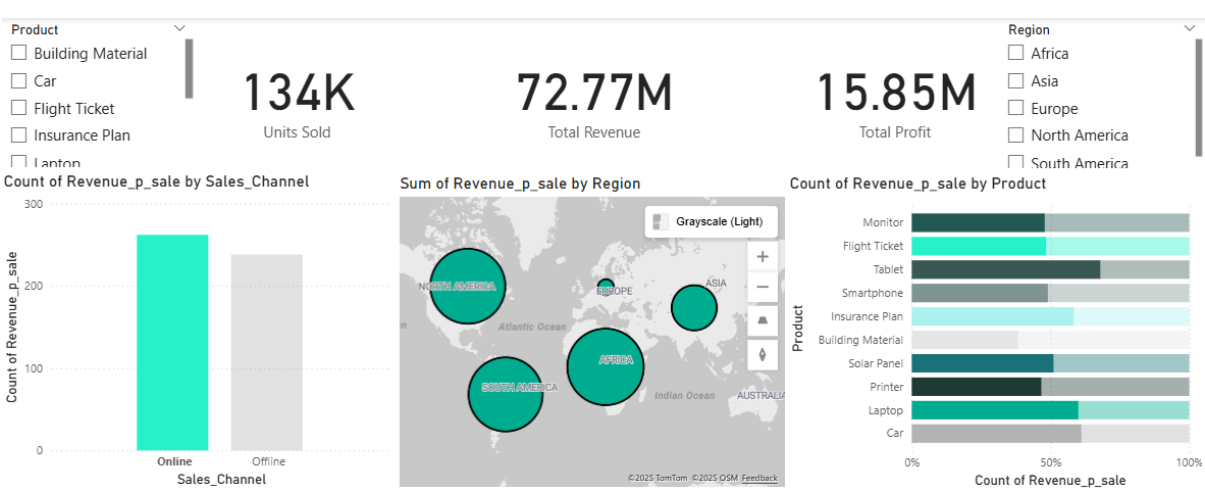
Indian Ocean

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Product

Count of Revenue_p_sale

Appendix 2.3



Region

☐ Africa

☐ Asia

☐ Europe

☐ North America

☐ South America

134K

Units Sold

72.77M

Total Revenue

15.85M

Total Profit

Count of Revenue_p_sale by Sales_Channel

Sum of Revenue_p_sale by Region

Count of Revenue_p_sale by Product

Count of Revenue_p_sale

Sales_Channel

Grayscale (Light)

NORTH AMERICA

SOUTH AMERICA

EUROPE

AFRICA

ASIA

AUSTRALIA

Atlantic Ocean

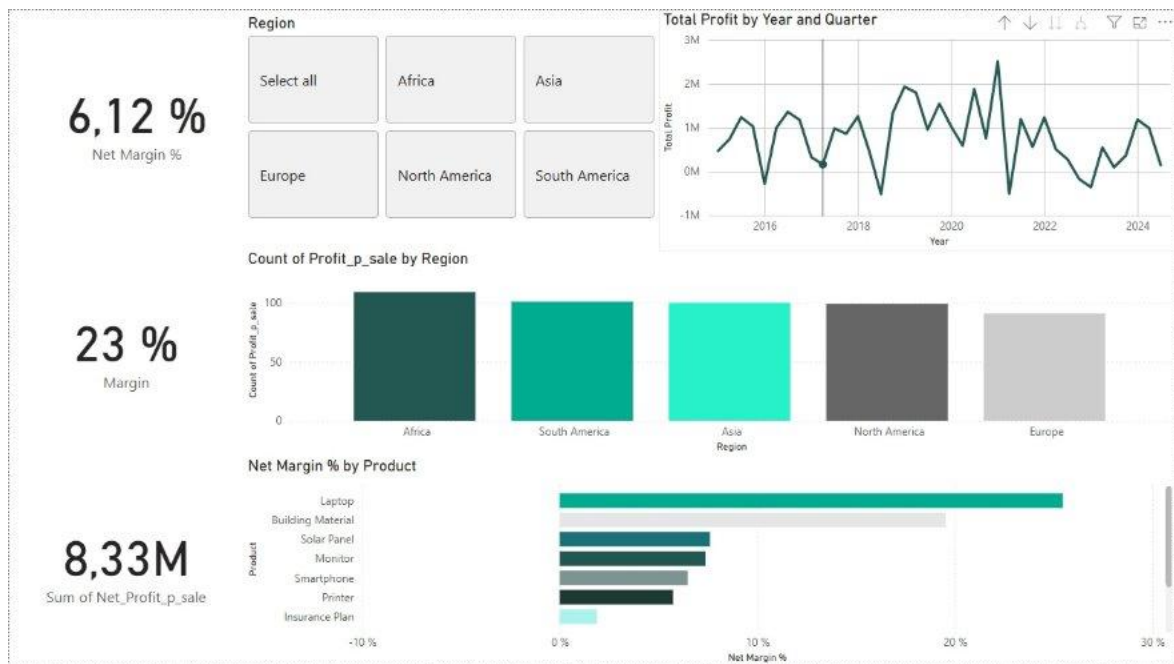
Indian Ocean

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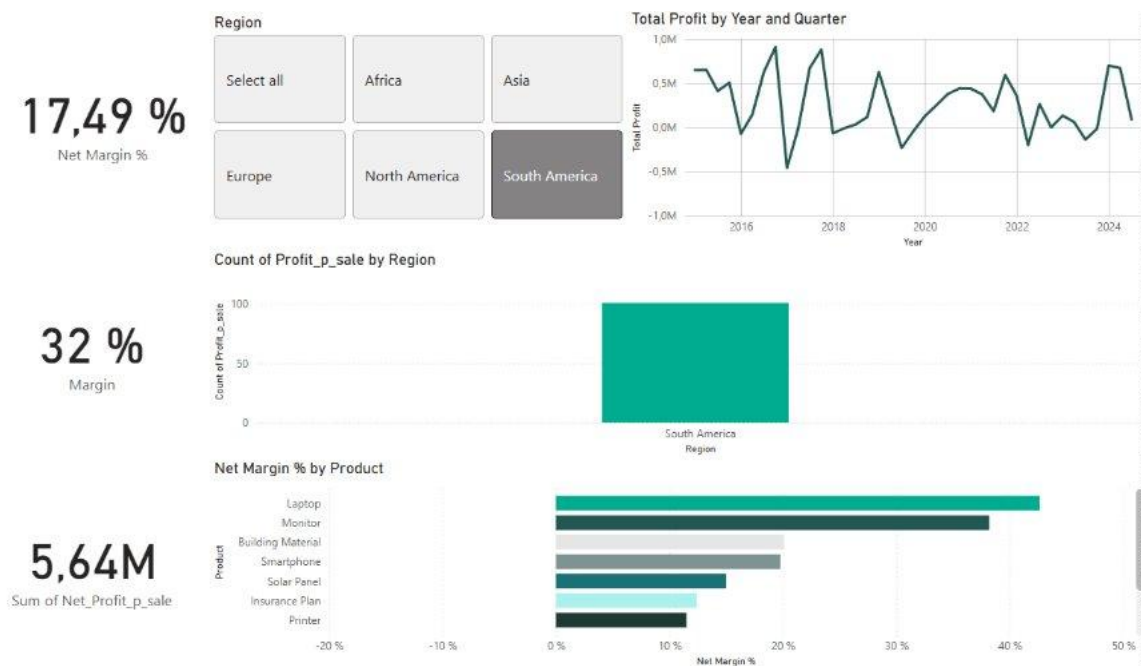
Product

Count of Revenue_p_sale

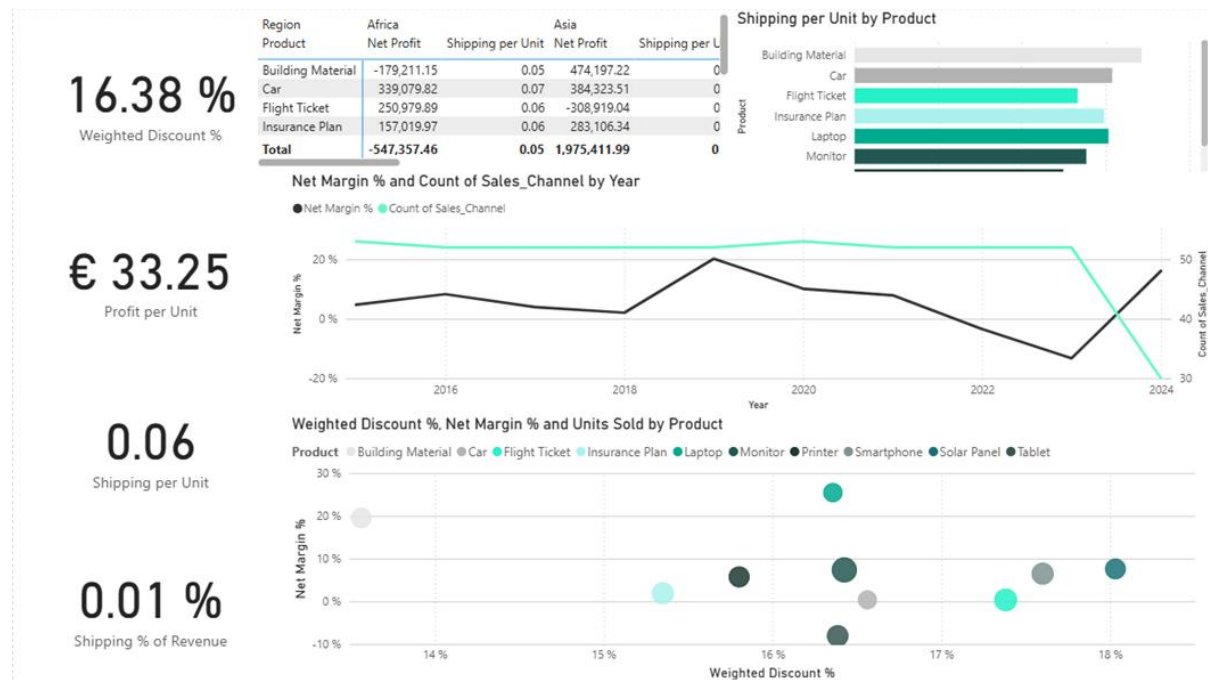
Appendix 3.1



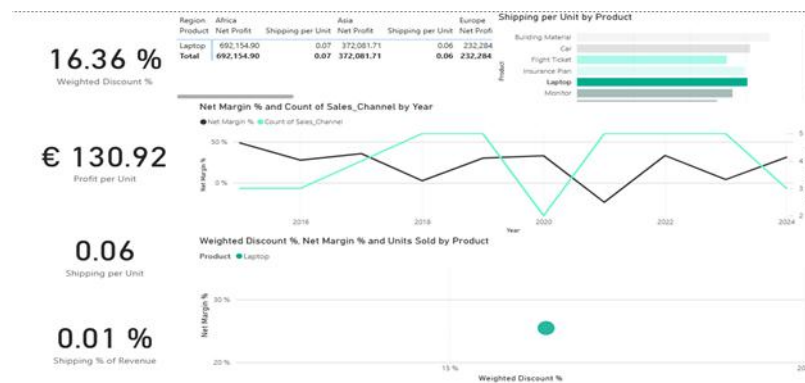
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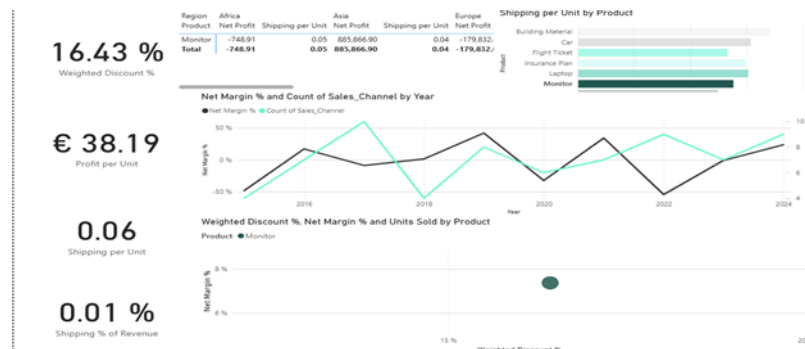
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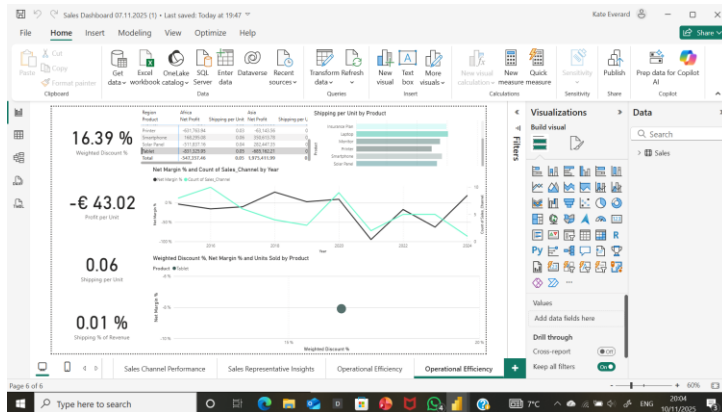
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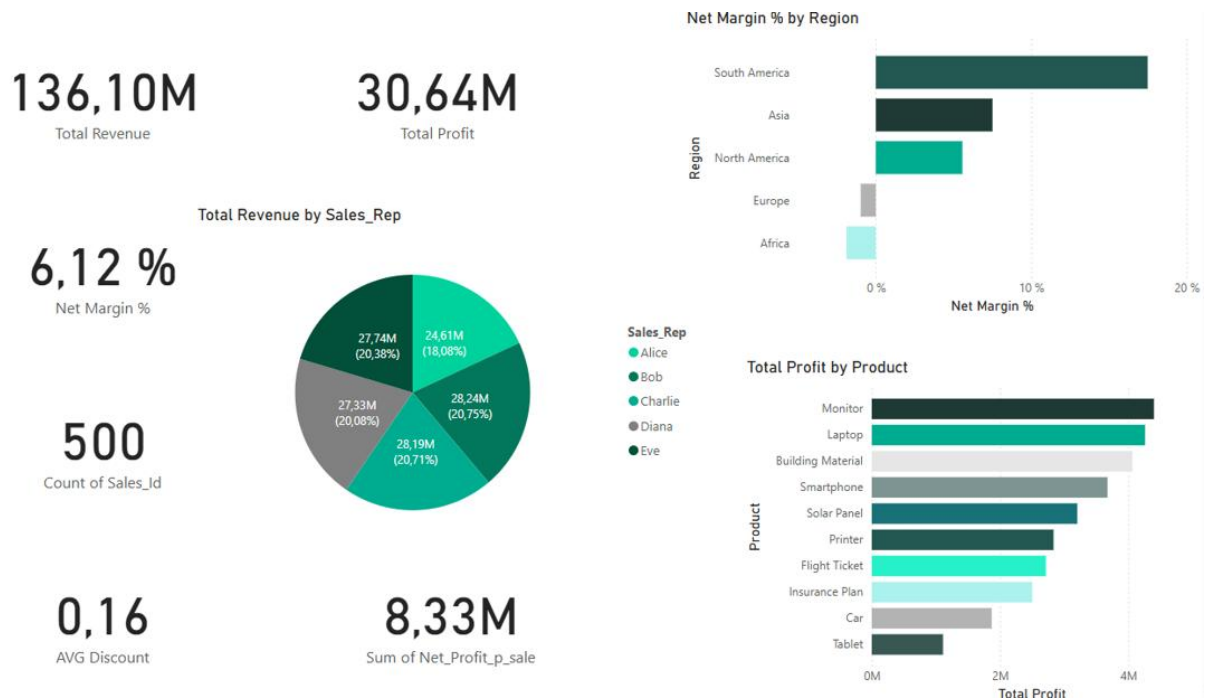
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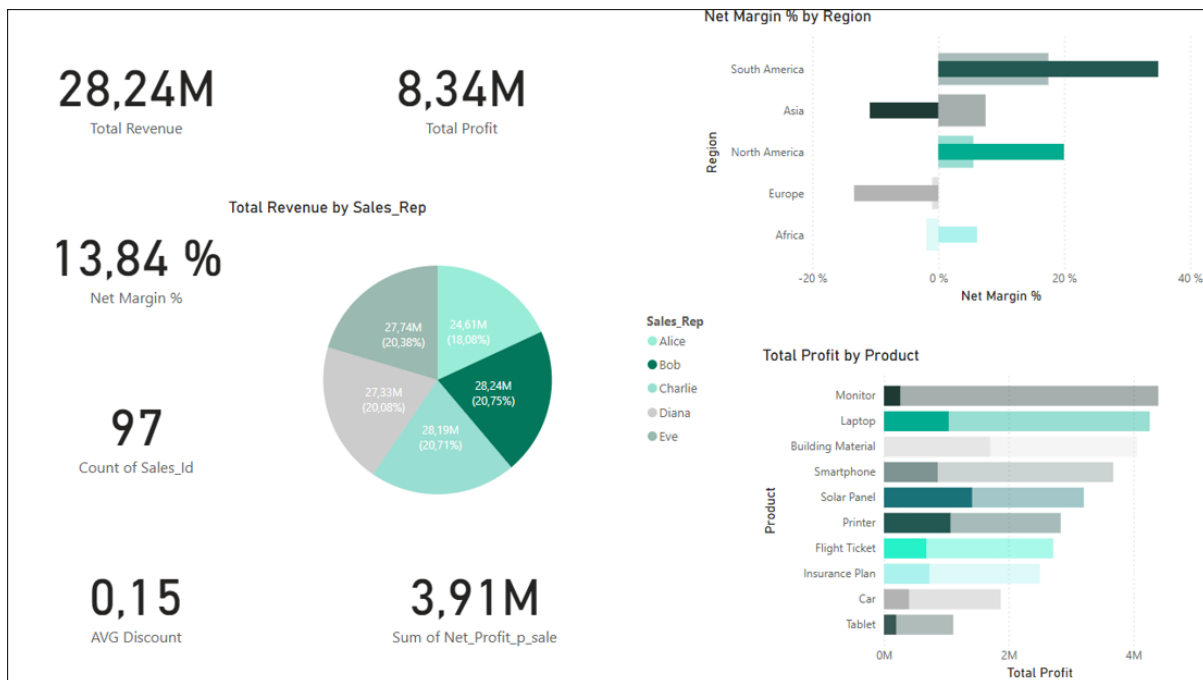
Appendix 4.4



Appendix 5.1



Appendix 5.2



Appendix 5.3

